

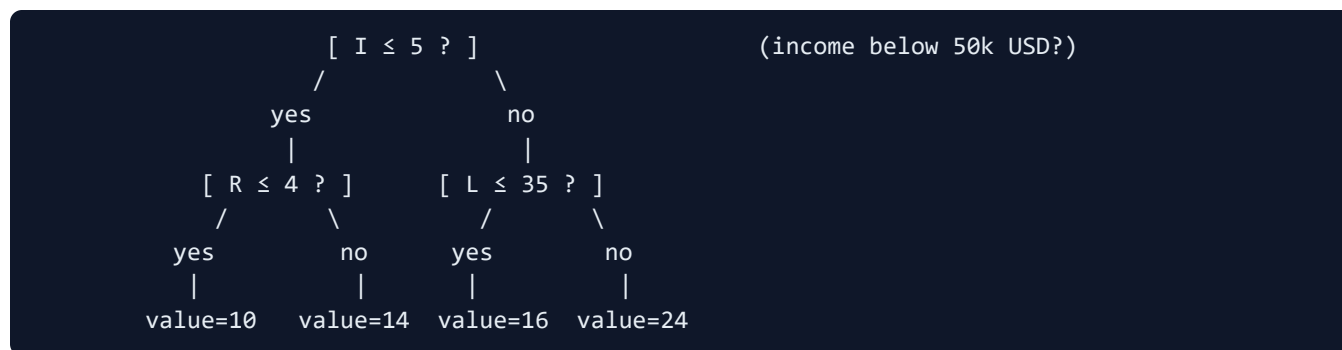
SHAP by hand (shallow tree) — worksheet

OPIM 5512 · Data Science Using Python · Dr. Dave Wanik — name: _____

SHAP explains **one** prediction: how much did each feature push it **above or below what we'd normally expect**? The "fair" split comes from **Shapley values** — a feature's *average marginal contribution* over every coalition it could join. Here you compute the whole thing by hand from a **printed decision tree**; you'll then recognize the exact same steps (on a real California-housing tree) in the notebook `2a_SHAP_ShallowTree_CAHousing`.

The model — one shallow tree

A tree scores a home (median value, in **units of 100,000 USD**) from three features: **I** = income (in units of 10,000 USD), **R** = rooms, **L** = latitude.



The value function — how we score a coalition

To score a coalition S (the features we treat as "**known**"), we set the features in S to **this home's** values and the rest to a **baseline** row, then read the tree's leaf.

Casual name for $v(S)$: call it "**the team's score**" — the price the tree predicts when only the *players* (features) in S are **on the field**, and everyone else sits at their baseline value.

feature	baseline row	this home x
I (income, 10k USD)	3	8
R (rooms)	3	6
L (latitude)	34	38

So $v(\emptyset)$ = the score with **nobody** on the field (all-baseline row) = the *starting* score, and $v(\{I, R, L\})$ = the score with **everyone** on (all-home row) = the real prediction.

How the "background" works for a real tree (this matters in the notebook; here we fake it with one baseline row). A *linear* model lets you plug in one "average house" and predict once. A **tree can't** — it's stepwise — so instead it **averages the tree's predictions over a whole set of reference houses (the *background*)**, and that average is the **base value**. `TreeExplainer` can even skip the extra data: each leaf already *remembers how many training houses landed in it*, so it weights the leaves by those counts to get the average automatically ("**path-dependent**"). Hand it a background sample instead and it simply averages `model.predict(background)` ("**interventional**"). Our single baseline row is the by-hand stand-in for that background.

The Shapley value (coalition form), for $p = 3$ features:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} w(|S|) [v(S \cup \{i\}) - v(S)], \quad w(|S|) = \frac{|S|! (p - |S| - 1)!}{p!}.$$

How to read that formula

One sentence: "*for every team feature i could join, measure i 's bump, weight it, and add them up.*"

Symbol by symbol — the **last column plugs in our problem** (with i = income):

symbol	casual meaning	our example (i = income)
N	all the features	$\{I, R, L\}$
$N \setminus \{i\}$	N without i (the $\setminus$ means "set minus")	$\{R, L\}$
$S \subseteq N \setminus \{i\}$	a team of the <i>other</i> features (the $\sum$ loops over all of them)	one of $\{\}, \{R\}, \{L\}, \{R, L\}$
$S \cup \{i\}$	union = the team S with i added in	$\{R\} \cup \{I\} = \{I, R\}$
$v(S \cup \{i\}) - v(S)$	team's score with i minus without = i 's bump	$v(\{I, R\}) - v(\{R\}) = 16 - 14 = 2$
$w(S)$	the weight (depends only on the team's size)	$\frac{1}{3}$ or $\frac{1}{6}$

Where the union $\cup$ comes in: it's just the " $+ i$ joins the team" step. You trace the tree once with team S known ($v(S)$, so i is on baseline), then **again** with S **and** i known ($v(S \cup \{i\})$) — the difference is what i added.

The weights are constant. $w(|S|)$ depends only on the team **size** (and p) — **not** on the house. For $p = 3$ they are **always** $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$, reused for every prediction. Only the $v(S)$ numbers change from row to row.

Tasks

1. **Trace the tree** to fill in the value of all 8 coalitions. For each, set the listed features to the home's value and the rest to baseline, then follow the tree to a leaf.

S	$\{\}$	$\{I\}$	$\{R\}$	$\{L\}$	$\{I, R\}$	$\{I, L\}$	$\{R, L\}$	$\{I, R, L\}$
$v(S)$								

2. **Weights.** Compute $w(|S|)$ for $|S| = 0, 1, 2$ (with $p = 3$).
 3. ϕ_I . List the four terms $w \cdot [v(S \cup \{I\}) - v(S)]$ for $S = \{\}, \{R\}, \{L\}, \{R, L\}$ and add them.
 4. ϕ_R and ϕ_L . Same for rooms and latitude.
 5. **Efficiency (local accuracy).** Check that **base value** + $\phi_I + \phi_R + \phi_L = \text{prediction}$.
 6. **Interpret the interactions.** Explain *from the tree* why $v(\{I, R\}) = v(\{I\})$ (does the tree ever look at R once income is high?), and why $v(\{L\}) = v(\emptyset)$.
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